

Probability

Random events happen by chance.

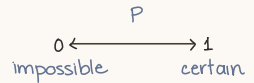
fair, unbiased

Probability is a measure of how likely they are.

1 The basics

- outcome / elementary event = the result of an experiment
- events = 'groups' of ≥ 1 outcomes
- when all outcomes are equally likely (equiprobable)
 - calculate probability: each repeat = a 'trial'

$$P(\text{event}) = \frac{\text{No. of outcomes when event happens}}{\text{Total no. of possible outcomes}}$$



EXAMPLE Suppose I've got a bag with 15 balls in — 5 red, 6 blue and 4 green.

If I take a ball out without looking, then any ball is equally likely — there are 15 possible outcomes. Of these 15 outcomes, 5 are red, 6 are blue and 4 are green. And so...

$$P(\text{red ball}) = \frac{5}{15} = \frac{1}{3}$$

$$P(\text{blue ball}) = \frac{6}{15} = \frac{2}{5}$$

$$P(\text{green ball}) = \frac{4}{15}$$

You can find the probability of either a red or a green ball in a similar way...

$$P(\text{red or green ball}) = \frac{9}{15} = \frac{3}{5}$$

- exhaustive events: contains all possible outcomes of an experiment
e.g.: $P(\text{red}) + P(\text{not red}) = 1$
- expectation: estimate likelihood of event in a series of trials based on probability

2 Mutually exclusive events, the addition law

cannot both occur at the same time.

e.g. Roll a die once. I can get a 1 **OR** a 5.
CAN'T get a 1 **AND** a 5.

$$P(1 \text{ or } 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

$$P(A) + P(A') = 1$$

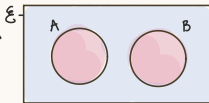
$$P(A \text{ or } B) = P(A) + P(B)$$

* Venn diagrams

if **mutually exclusive**

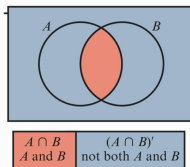
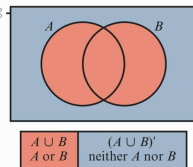
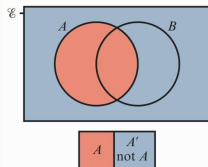
no overlap: $P(A \cap B) = 0$

$A \cap B = \emptyset$ empty set



then:

$$P(A \cup B) = P(A) + P(B)$$



A and B overlaps
= has common favorable outcome
= **NOT** mutually exclusive

3. Independent events; the multiplication law

occur with **NO** effect on each other → the other can occur or not occur, doesn't matter.

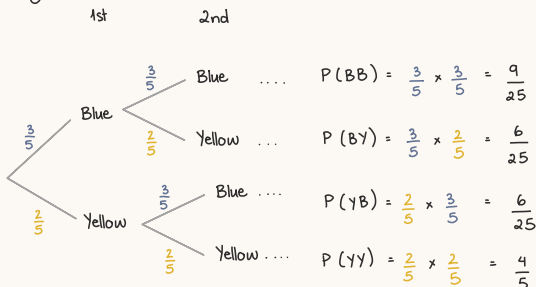
e.g.: A bag has 3 blue marbles and 2 yellow marbles.

Pick 1, record color. **REPLACE** it. Pick another.

ensures there are always 5 marbles to pick from.

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)$$

* Tree diagrams



4. Conditional probability

A child is selected at random from a group of 11 boys and nine girls, and one of the girls is called Rose. Find the probability that Rose is selected, given that a girl is selected.

Answer

$$P(\text{Rose is selected} \mid \text{a girl is selected}) = \frac{1}{9}$$

The additional information, 'given that a girl is selected', reduces the number of possible selections from 20 to 9, and Rose is one of those nine girls.

probability of A,
given that B has
already happened

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

5. Dependent events and conditional probability

→ occurrence of 1 affects the other

e.g.: making a selection without replacement

→ 2nd outcome depends on 1st outcome

$$\begin{aligned} P(A \text{ and } B) &= P(A \cap B) = P(A) \times P(B \mid A) \\ P(B \text{ and } A) &= P(B \cap A) = P(B) \times P(A \mid B) \end{aligned}$$

$$P(A) \times P(B \mid A) = P(B) \times P(A \mid B)$$

* if B is independent of A:

$$P(B \mid A) = P(B \mid A') = P(B)$$

probability that B occurs, given A occurs = probability that B occurs given A doesn't occur

→ A and B are not affected by each other